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Problems to be discussed:

Comments:

Homework attached on page:

6. Two concentric circles are centered at O(-4, 3). The ratio of the area of inner circle to that of the outer circle is 1:9. Points A and B lie on the boundaries of the inner and outer circle, respectively, as shown below.

The coordinates of point B are (3, 5).

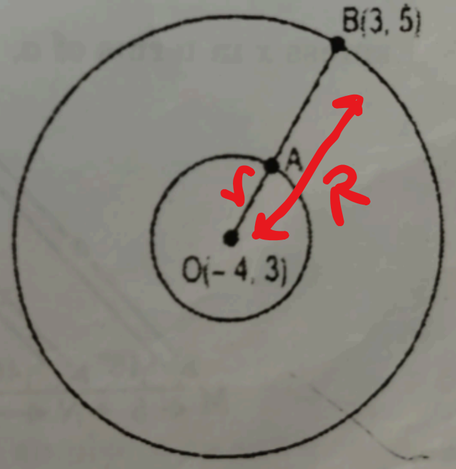
Which of the following are the coordinates of A?

a) $\left(\frac{2}{3}, \frac{13}{3}\right)$

b) $\left(\frac{-5}{3}, \frac{11}{3}\right)$

c) $\left(\frac{-9}{4}, \frac{14}{4}\right)$

d) $\left(\frac{-33}{10}, \frac{16}{5}\right)$



$\frac{r}{R} = \frac{1}{3}$

$\frac{1}{9}$

m : n

1 : 2

O (-4, 3) A (x, y) B (3, 5)

$x = \frac{3 - 8}{3} = \frac{-5}{3}$

$y = \frac{11}{3}$

If $\operatorname{cosec}\theta = x + \frac{1}{4x}$, prove that $\operatorname{cosec}\theta + \cot\theta = 2x$ or $\frac{1}{2x}$.

- $\sin^2\theta + \cos^2\theta = 1$
- $\sec^2\theta - \tan^2\theta = 1$
- $\operatorname{cosec}^2\theta - \cot^2\theta = 1$

$$\cot^2\theta = \operatorname{cosec}^2\theta - 1$$

$$= \left(x + \frac{1}{4x}\right)^2 - 1$$

$$= x^2 + \frac{1}{16x^2} + \frac{1}{2} - 1$$

$$= x^2 + \frac{1}{16x^2} - 2 \cdot \frac{1}{4x} \cdot x$$

$$\cot^2\theta = \left(x - \frac{1}{4x}\right)^2$$

$$\cot\theta = x - \frac{1}{4x} \quad ; \quad \frac{1}{4x} - x$$

$x^2 = 9$

$x = 3, -3$

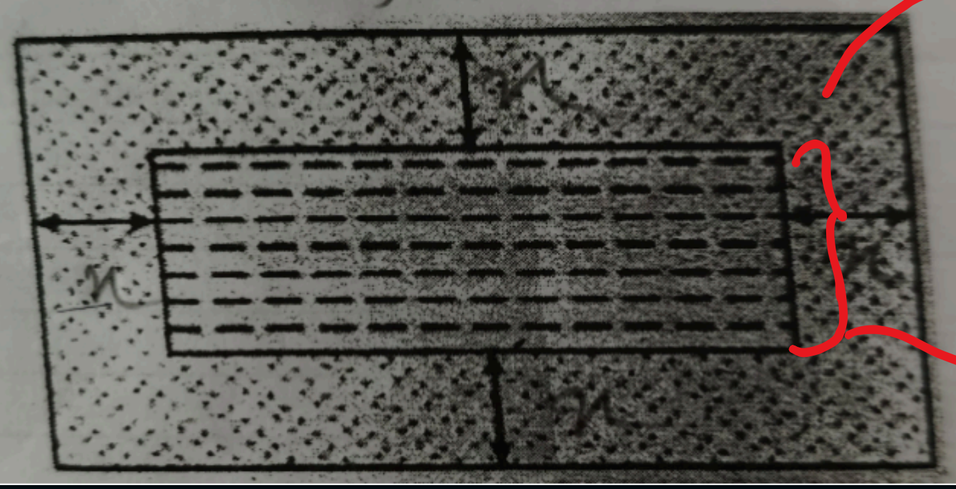
$\operatorname{cosec}\theta + \cot\theta$

$= 2x$ or $\frac{1}{2x}$

In the centre of a rectangular lawn of dimensions 50 m × 40 m, a rectangular pond has to be constructed, so that the area of the grass surrounding the pond would be 1184 m². Find the length and breadth of the pond.

=

[11]



$(50-2x)(40-2x) = 816$

$4x^2 - 180x + 2000 = 816$

$\rightarrow 4x^2 - 180x + 1184 = 0$

$x^2 - 45x + 296 = 0$

$x = \frac{45 \pm \sqrt{2025 - 1184}}{2}$

$= \frac{45 \pm 29}{2}$

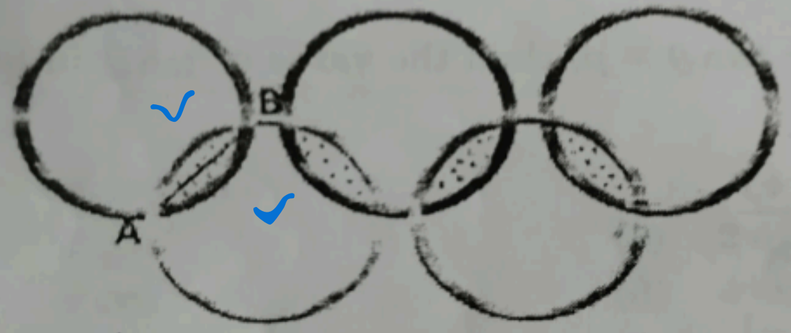
$= 37$ or 8

$50 - 2x = 34$ m

$40 - 2x = 24$ m

$\Rightarrow 34 \times 24$ m

Given below is the picture of the Olympic rings made by taking five congruent circles of radius 1cm each, intersecting in such a way that the chord formed by joining the point of intersection of two circles is also of length 1cm. Total area of all the dotted regions assuming the thickness of the rings to be negligible is :

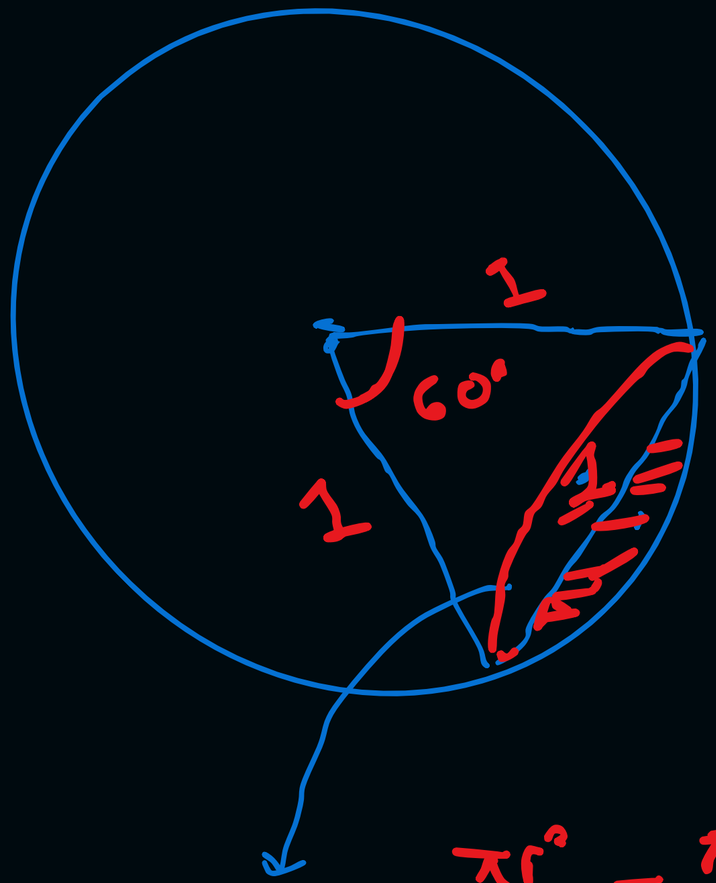


(a) $4\left(\frac{\pi}{12}-\frac{\sqrt{3}}{4}\right)$ cm²

(b) $\left(\frac{\pi}{6}-\frac{\sqrt{3}}{4}\right)$ cm²

(c) $4\left(\frac{\pi}{6}-\frac{\sqrt{3}}{4}\right)$ cm²

(d) $8\left(\frac{\pi}{6}-\frac{\sqrt{3}}{4}\right)$ cm²



$\frac{\pi r^2}{6} = \frac{\pi \times 1}{6} = \frac{\pi}{6}$

$-\frac{\sqrt{3}}{4}$]

If $(2\sin\theta + 3\cos\theta) = 2$, prove that $(3\sin\theta - 2\cos\theta) = \pm 3$.

$9\sin^2\theta + 9\cos^2\theta + 12\sin\theta\cos\theta = 4$ $\quad \vec{4} = c$

$9\sin^2\theta + 4\cos^2\theta - 12\sin\theta\cos\theta = c^2$

$13\sin^2\theta + 13\cos^2\theta = 4 + c^2$

$13 = 4 + c^2$

$c^2 = 9$

$c = \pm 3$ ✓ [QED]

If $\tan A = n \tan B$ and $\sin A = m \sin B$, prove that $\cos^2 A = \dfrac{(m^2-1)}{(n^2-1)}$.

$$\frac{\sin A}{\cos A} = \frac{n \sin B}{\cos B} \Big/ m \sin B$$

$$\cos A = \frac{m}{n} \cos B$$

$$\frac{n^2 \cos^2 A}{m^2} = \cos^2 B \quad \text{— (1)}$$

$$\frac{1}{\cos A} = \frac{n}{m \cos B}$$

$$\sin^2 A = m^2 \sin^2 B$$

$$m \cos B = n \cos A$$

$$(1 - \cos^2 A) = m^2 (1 - \cos^2 B)$$

$$1 - \cos^2 A = m^2 - m^2 \cos^2 B \quad \text{— (2)}$$

[Subs. (1) in (2)]

$$1 - \cos^2 A = m^2 - m^2 \times \frac{n^2}{m^2} \cos^2 A$$

$$1 - \cos^2 A = m^2 - n^2 \cos^2 A$$

$$1 - m^2 = (1 - n^2) \cos^2 A \Rightarrow \cos^2 A = \frac{1-m^2}{1-n^2} = \boxed{\frac{m^2-1}{n^2-1}}$$

If $\tan A = n \tan B$ and $\sin A = m \sin B$, prove that $\cos^2 A = \frac{(m^2-1)}{(n^2-1)}$.

$\sin^2 A = m^2 \sin^2 B$

$+ \quad n^2 \cos^2 A = m^2 \cos^2 B$

$n^2 \cos^2 A = m^2$

$+ 1 - \cos^2 A$

$n \cos A = m \cos B$

$(n^2 - 1)\cos^2 A + 1 = m^2$

$(n^2 - 1)\cos^2 A = m^2 - 1$

$\cos^2 A = \frac{m^2 - 1}{n^2 - 1}$

If the ratio of the sums of n terms of two APs is (7n + 1) : (4n + 27), find the ratio of their (i) mth terms and (ii) 9th terms.

AP → $S_n = \frac{n}{2}[2a + (n-1)d]$

$S_n' = \frac{n}{2}[2A + (n-1)D]$

$$\frac{S_n}{S_n'} = \frac{7n+1}{4n+27} = \frac{2a + (n-1)d}{2A + (n-1)D} \quad \begin{matrix} \to (7n+1)k \\ \to (4n+27)k \end{matrix}$$

$2\left(\frac{a}{k}\right) + (n-1)\left(\frac{d}{k}\right) = (7n+1)$

$\left(\frac{d}{k}\right)n + \left(2\frac{a}{k} - \frac{d}{k}\right) = 7n + 1$

$\frac{d}{k} = 7$

$2\frac{a}{k} - \frac{d}{k} = 1 \Rightarrow \frac{a}{k} = 4$

If the ratio of the sums of n terms of two APs is $(7n + 1) : (4n + 27)$, find the ratio of their (i) mth terms and (ii) 9th terms.

$$\frac{2A + (n-1)D}{} = (4n + 27)k$$

$$2\frac{A}{k} + (n-1)\frac{D}{k} = 4n + 27$$

$$\Rightarrow \left(\frac{D}{k}\right)n + \left(2\frac{A}{k} - \frac{D}{k}\right) = 4n + 27$$

$$\Rightarrow \frac{D}{k} = 4 \quad \& \quad \frac{2A}{k} - \frac{D}{k} = 27 \Rightarrow \frac{A}{k} = \frac{31}{2}$$

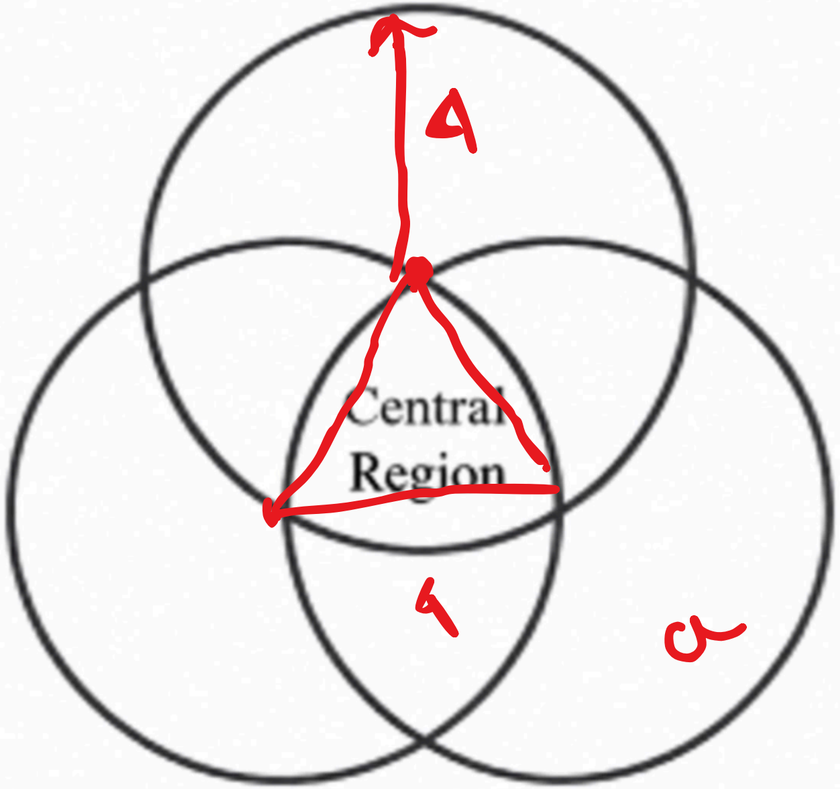
$$\frac{T_m}{T_{m'}} = \frac{a + (m-1)d}{A + (m-1)D} = \frac{\frac{a}{k} + (m-1)\frac{d}{k}}{\frac{A}{k} + (m-1)\frac{D}{k}} = \frac{4 + (m-1)7}{\frac{31}{2} + (m-1)4}$$

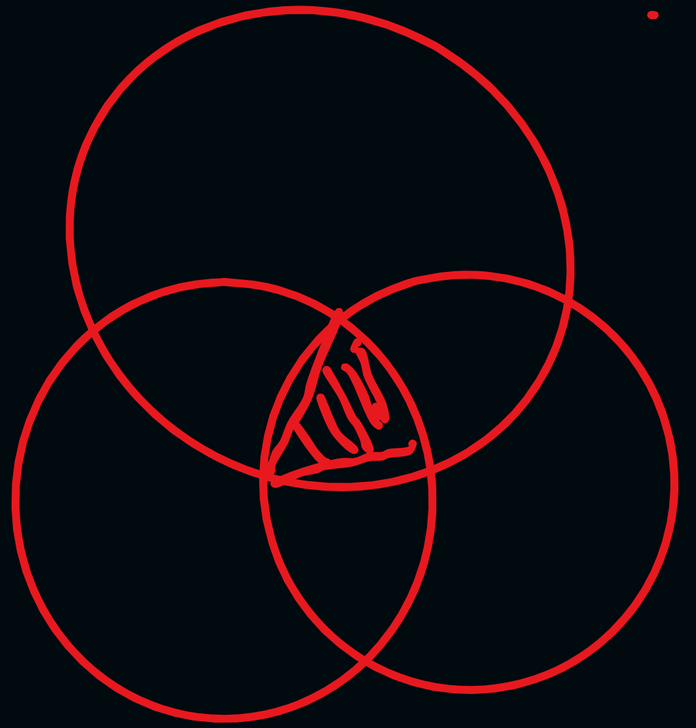
$$= \frac{7m - 3}{4m + \frac{23}{2}}$$

$$= \frac{19m - 6}{8m + 23}$$

$$\therefore = \frac{120}{95} = \frac{24}{19}$$

Find out the area of the central region.





$\Delta + 3a \rightarrow 3\left(\frac{\pi \times 4^2}{6} - \frac{\sqrt{3}}{4} \times 4^2\right)$

$4\sqrt{3} + (8\pi - 12\sqrt{3}) = 8\pi - 8\sqrt{3}$

$8(\pi - \sqrt{3})$



Sector − △

$\frac{\pi r^2}{6} - \frac{\sqrt{3}}{4}r^2$

$a = \frac{r^2}{2}\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)$

$\hookrightarrow 3a = \frac{r^2}{2}\left(\pi - \frac{3\sqrt{3}}{2}\right)$

$\hookrightarrow \Delta + 3a = \frac{\sqrt{3}}{4}r^2 + \frac{r^2}{2}\left(\pi - \frac{3\sqrt{3}}{2}\right)$

$= \frac{r^2}{2}\left(\pi - \frac{3\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right)$

$= \frac{r^2}{2}(\pi - \sqrt{3})$

In our qsn, $r = 4$

$\Rightarrow \frac{16}{2}(\pi - \sqrt{3})$

$= 8(\pi - \sqrt{3})$

If $a_k = \sin^k\theta + \cos^k\theta$, for all real values of k, and the value of $\left(\frac{a_7 - a_9}{a_5}\right) = \frac{2p}{(1+\cot^2\theta)(1+\tan^2\theta)}$, then value of 'p' is

- ○ 1
- ○ 2
- ○ $\frac{1}{2}$ ✓
- ○ 0

$(1+\cot^2\theta)(1+\tan^2\theta) = \operatorname{cosec}^2\theta \sec^2\theta$

$2p\sin^2\theta\cos^2\theta$

$a_7 - a_9$

$= \sin^7\theta + \cos^7\theta - \sin^9\theta - \cos^9\theta$

$= (\sin^7\theta - \sin^9\theta) + (\cos^7\theta - \cos^9\theta)$

$= \sin^7\theta(1-\sin^2\theta) + \cos^7\theta(1-\cos^2\theta)$

$= \sin^7\theta\cos^2\theta + \cos^7\theta\sin^2\theta$

$= \sin^2\theta\cos^2\theta(\sin^5\theta + \cos^5\theta)$

$= \sin^2\theta\cos^2\theta\, a_5$

$\Rightarrow LHS = \sin^2\theta\cos^2\theta$

$RHS = 2p\sin^2\theta\cos^2\theta$

$\Rightarrow 2p = 1 \Rightarrow p = \frac{1}{2}$

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